

Carbon Dioxide Hydrogenation To Form Methanol via a Reverse-Water-Gas-Shift Reaction (the CAMERE Process)

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The CAMERE process (carbon dioxide hydrogenation to form methanol via a reverse-water-gas-shift reaction) was developed and evaluated. The reverse-water-gas-shift reactor and the methanol synthesis reactor were serially aligned to form methanol from CO₂ hydrogenation. Carbon dioxide was converted to CO and water by the reverse-water-gas-shift reaction (RWReaction) to remove water before methanol was synthesized. With the elimination of water by RWReaction, the purge gas volume was minimized as the recycle gas volume decreased. Because of the minimum purge gas loss by the pretreatment of RWReactor, the overall methanol yield increased up to 89% from 69%. An active and stable catalyst with the composition of Cu/ZnO/ZrO₂/Ga₂O₃ (5:3:1:1) was developed. The system was optimized and compared with the commercial methanol synthesis processes from natural gas and coal.

1. Introduction

Hydrogenation of carbon dioxide was one of the major approaches for CO₂ reduction to mitigate the greenhouse effects. The formation of oxygenated compounds, such as methanol, ethanol, and formic acid, has been intensively investigated in view of their added value. Especially the formation of methanol from the hydrogenation of CO₂ has drawn much attention because of its minimum hydrogen requirement and its large demand as a bulk chemical. Further, methanol is a key material for C₁ chemistry.

Commercially, methanol was synthesized from syngas prepared from coal or natural gas. The reaction feed of the conventional process consists of not only CO and H₂ but also 10% CO₂ as one of the main components.¹ It has been reported that CO₂ is the carbon source for methanol synthesis.² Accordingly, there have been tremendous efforts to develop a catalytic system for the CO₂ hydrogenation.^{3–6}

The element copper was a major component of both CO₂ and CO hydrogenation catalysts. Nevertheless, the activity of the copper-containing catalysts was suppressed with pure CO₂/H₂ feed. One of the major reasons for the catalyst activity suppression was the presence of water. The water, which was formed from the CO₂ hydrogenation, adsorbed on the active site of the copper catalyst and inhibited the adsorption of CO₂ for the next catalytic reactions. When carbon monoxide is present in the reaction mixture, it reacts with water to form CO₂ and hydrogen to regenerate active sites on copper. Carbon monoxide is a good water scavenger.⁷

Thus, the elimination of water from the reaction system turned out to be a key issue improving the

catalytic activity and the system efficiency. German engineering company Lurgi Co. reported a two-methanol-reactor system for the methanol synthesis from CO₂/H₂ feed.⁸ Two methanol reactors were placed in a row. The output of the first reactor was introduced into the rectifier of the second reactor to eliminate water and methanol. They claimed improved efficiency for the production of methanol. However, the formation of water and CO from the CO₂/H₂ mixture depends on temperature, and it is favored at high temperature. The temperature of the first methanol reactor should be kept at no more than 573 K to maintain the thermal stability of the copper catalyst. CO₂ could not be sufficiently converted into CO and H₂O at this temperature.

Herein, we report the development of the CAMERE process (carbon dioxide hydrogenation to form methanol via a reverse-water-gas-shift reaction). The proper copper-containing catalysts for the high CO₂ concentration feed were developed, and the system optimization was also performed. The process characteristics were compared with those of the direct CO₂ hydrogenation process and the commercial methanol synthesis processes from syngas.

2. Experimental Section

Methanol synthesis catalysts were prepared by a coprecipitation of the corresponding metal nitrates.⁹ The composition of the catalysts was controlled in molar ratios. Catalysts were reduced in situ under hydrogen (5% in helium) at 573 K for 1 h before the reaction. The reaction was performed under the reaction conditions of 27.2 atm and GHSV of 12 000. The feed composition was 40% CO₂, and the ratio of hydrogen to carbon source (H₂/CO₂ + CO) was 2.4. Two commercial methanol synthesis catalysts were examined under the same reaction conditions described above.

A minipilot operation was performed: a tubular catalytic reactor was equipped with the gas recycle system and operated with a commercial methanol syn-

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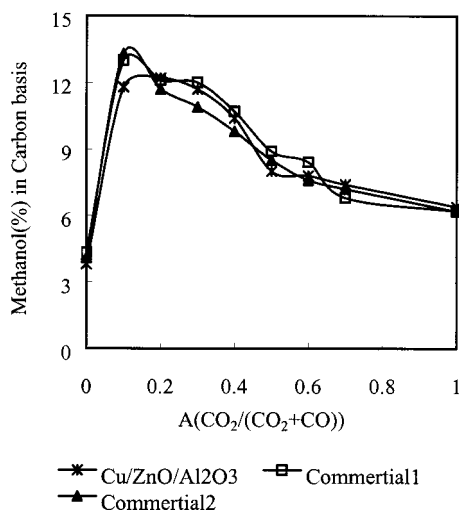
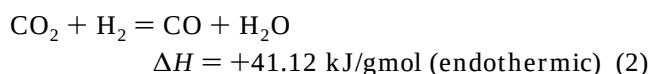
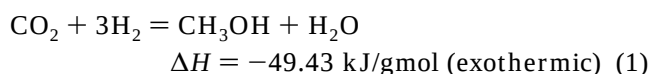


Figure 1. Methanol yields per pass depending on the A value. Reaction conditions: 523 K, 27.2 atm, GHSV of 19 200, $H_2/(CO_2 + CO) = 3$.

thesis catalyst (15–20 mesh, catalyst porosity $0.36 \text{ cm}^2/\text{g}$, specific surface area $81 \text{ m}^2/\text{g}$). The absence of diffusion limitations was confirmed by changing catalyst sizes, temperatures, and flow rates.

3. Results

3.1. Methanol Yield per Pass vs CO_2 Concentration in the Reaction Feed. Two major reactions are involved in the CO_2 hydrogenation to give methanol.



The two reactions compete with each other in the course of the CO_2 hydrogenation. Methanol formation is thermodynamically favored under the reaction conditions of low temperature and high pressure, whereas CO formation is favored at high temperature near equilibrium conditions.¹⁰

As mentioned above, the formation of water blocked the active sites and reduced the catalyst activity. The presence of CO in the reaction mixture eliminated water from the copper surface by RWReaction and recovered the catalyst activities. Therefore, the concentration of CO in the reaction mixture was a critical parameter for the high methanol.

Methanol yields per pass were examined according to the ratio of $CO_2/(CO_2 + CO)$ for three different catalysts. Cu/ZnO/ Al_2O_3 (6:3:1) and two commercial catalysts were tested and compared to each other. The results were summarized in Figure 1. The A value in Figure 1 means the ratio of CO_2 to the total moles of CO_2 and CO in the feed gas.

The methanol yield per pass was dependent on the CO_2 concentration in the reaction feed. When CO_2 was only injected ($A = 1$) into the methanol synthesis reactor as a carbon source, the methanol yield per pass was about 6% on a carbon basis. As the CO_2 concentration decreased (40%, $A = 0.4$), the methanol yield per pass increased to 10%. The increment was continued up to 13.4% as the CO_2 concentration decreased down to 10% ($A = 0.1$). Meanwhile, the methanol yield per pass

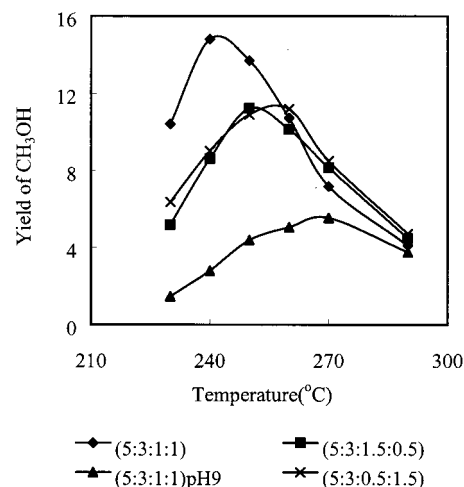


Figure 2. Methanol yields per pass with different compositions of CuO/ZnO/ZrO₂/Ga₂O₃ catalyst depending on the reaction temperature. Reaction conditions: 27.2 atm, GHSV of 12 000, $CO_2/(CO_2 + CO) = 0.4$, $H_2/(CO_2 + CO) = 2.4$.

rapidly decreased to 4% with CO only ($A = 0$) in the reaction feed. Those three catalysts showed similar performance depending on the CO_2 concentration in the feed gas. This result led us to adopt RWReaction for the fine-tuning of the feed gas composition.

3.2. Development of Cu/ZnO/ZrO₂/Ga₂O₃ Catalysts for the High CO_2 Concentration Feed. A new type of highly active copper-containing catalyst was developed for the high CO_2 concentration. The commercial catalysts of Cu/ZnO/ Al_2O_3 (6:3:1) showed a poor catalytic methanol yield per pass as well as durability at the high CO_2 concentration. Among the several catalysts investigated, Cu/ZnO/ZrO₂/Ga₂O₃ (5:3:1:1) was one of the best systems and produced a higher yield of methanol than those of other commercial catalysts. The presence of gallium promoted the stability as well as the reactivity of the catalytic system.

The methanol yield per pass was monitored with the mole ratio of the Cu/ZnO/ZrO₂/Ga₂O₃ catalyst (Figure 2). The catalysts prepared at pH 7 were active, and the optimum composition was 5:3:1:1, giving 15% of the methanol yield per pass at 513 K.

The catalyst performance was further compared to those of the two commercial catalysts under the same reaction conditions (Figure 3). The Cu/ZnO/ZrO₂/Ga₂O₃ (5:3:1:1) catalyst showed a higher methanol yield per pass, especially at the low temperature of 513 K, than those of the well-known commercial catalysts. The catalyst activity at low temperature is one of the important parameters for the catalyst development because the methanol formation is an exothermic reaction and its equilibrium constant increased at low temperature. The Cu/ZnO/ZrO₂/Ga₂O₃ (5:3:1:1) system developed in our laboratory exhibited a good methanol yield per pass at low temperature and high CO_2 concentration. The durability test was carried out for up to 70 h under the same reaction conditions (Figure 4). There was no noticeable deactivation during the reaction time.

3.3. Combination of RWReaction with the Methanol Synthesis Reaction (the CAMERE Process). As shown in Figure 1, the initial concentration of CO_2 in the reaction feed was a key factor for the high methanol yield. The methanol synthesis reactor was serially aligned after the RWReactor, as shown in Scheme 1.

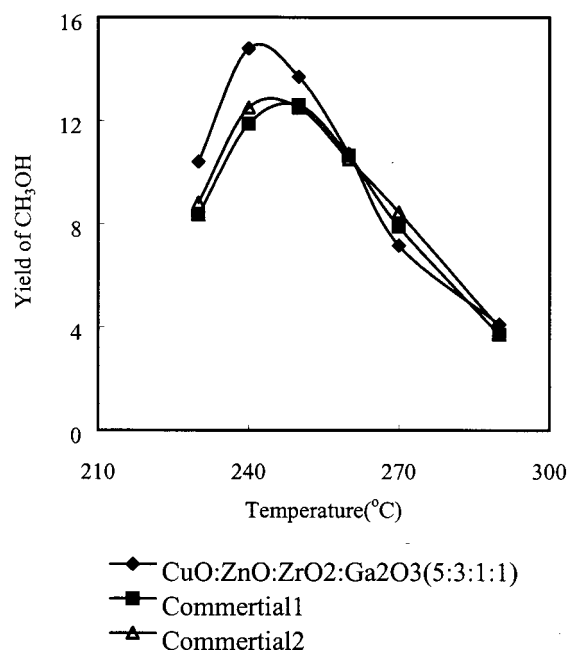


Figure 3. Comparison of the catalysts' performance. Reaction conditions: 27.2 atm, GHSV of 12 000, $\text{CO}_2/(\text{CO}_2 + \text{CO}) = 0.4$, $\text{H}_2/(\text{CO}_2 + \text{CO}) = 2.4$.

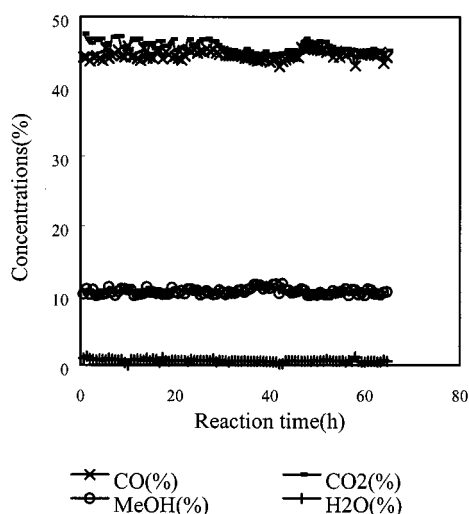


Figure 4. Durability test of $\text{CuO/ZnO/ZrO}_2/\text{Ga}_2\text{O}_3$ (5:3:1:1). Reaction conditions: 533 K, 27.2 atm, GHSV of 12 000, $\text{CO}_2/(\text{CO}_2 + \text{CO}) = 0.4$, $\text{H}_2/(\text{CO}_2 + \text{CO}) = 2.4$.

The CAMERE process was examined and compared with the direct hydrogenation process through the minipilot-plant operation. The results are summarized in Table 1. The data were recalculated based on the 300 000 ton/year commercial scale.

At the RWReactor, the CO_2 conversion to CO was 61%/pass, and the CO_2 composition in the outlet gas became 39% in carbon base. After the removal of water, the gas mixture was fed into the inlet of the methanol synthesis reactor.

One of the major differences in the methanol synthesis reactor was the reduction of the recycle gas volume. In the conventional direct methanol synthesis process from syngas, the methanol yield per pass was less than 10%. The rest of the unreacted gases should be recycled in order not to waste valuable hydrogen. Meanwhile, the commercial catalysts generally contained ppm levels of transition-metal impurities, which hydrogenated CO_2 into methane. As the recycle continued, the inert

methane partial pressure increased. To maintain the proper partial pressure of the feed gas, a portion of the recycle gas should be purged out. It was very important to notice that the gas purge from the reaction system meant the loss of the valuable hydrogen. The loss of hydrogen eventually decreased the overall methanol yield. The CAMERE process decreased the purge gas volume as the recycle gas volume decreased from 620 000 to 506 000 m^3/h . As a result, the overall methanol production increased from 37.01 to 47.87 ton/h, which was a 29% increase from the same amount of hydrogen.

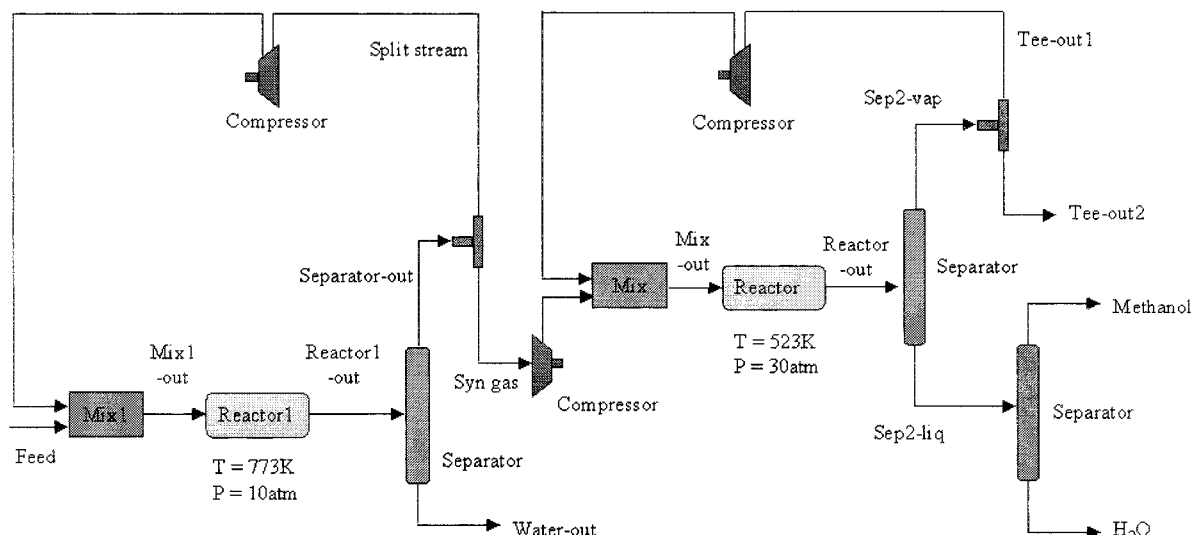
3.4. Equipment and Operation Costs of the CAMERE Process. System optimization was performed to minimize the equipment and the operation costs. Major variables for the CO_2 conversion to CO and water were temperature and the recycle ratio. The A values ($\text{CO}_2/(\text{CO}_2 + \text{CO})$) of output gas from the RWReactor were calculated by changing the temperature, recycle ratio, and pressure (Table 2). As the temperature increased, the CO_2 conversion to CO increased because of the endothermic nature of the reverse-water-gas-shift reaction. The CO_2 conversion per pass reached around 60% at 873 K, which was good enough for the next methanol synthesis. The overall CO_2 conversion to CO also increased with the increment of the recycle ratio. Because of the small entropy change, the CO_2 conversion was not affected by the pressure change. The feed composition of the methanol synthesis reactor was optimized on the basis of the operation cost, equipment cost, and overall methanol yield.

The CAMERA process diagram is depicted in Scheme 1. There were two main reactors combined sequentially. The downstream gases of the each reactor were recycled, and H_2O and $\text{H}_2\text{O}/\text{methanol}$ produced in each reactor were separated from the output gas streams just before being recycled. The reaction conditions of the RWReactor were 773 K and 10 atm, and those of the methanol reactor were 523 K and 30 atm. The material balance is summarized in Table 3.

Compared to the Lurgi process,⁸ the temperature of the RWReactor of the CAMERE process was easily elevated to 873 K to convert 60% of CO_2 . The recycle volume of the methanol synthesis of the CAMERE process was reduced, improving the process efficiency. The equipment and operation costs of the CAMERE process were calculated and compared with those of the conventional methanol synthesis processes from coal and natural gas on a 300 000 ton/year basis. The results are summarized in Table 4.

The capital cost of a reformer to prepare syngas from natural gas generally took a large part of the total investment, and more than half of the methanol production cost was attributed to the reformer price. The cost of syngas production from coal was even larger than that from natural gas because of the expensive coal gasifier. On the contrary, the CAMERE process did not require a reformer or a gasifier and, as a result, needed the low investment cost. The operation cost of the CAMERE process was a bit more than that of the natural gas process because of the increment of the recycle volume and the operation of two reactors but less than the other processes so far commercialized. To calculate the methanol production cost, the material costs of CO_2 and H_2 are needed along with the operation and investment costs. The raw material cost of CO_2 is dependent on the separation methods and CO_2 production sources. One

Scheme 1. Diagram of the CAMERE Process

Table 1. Comparison of the CAMERE Process with the Direct CO₂ Hydrogenation

	direct	CAMERE process
1. Reverse-Water-Gas-Shift Reaction		
catalyst, m ³		30
pressure, atm		20
feed gas, m ³ /h		150 000
space velocity, h ⁻¹		5000
condensed water, tons/h		18.5
conversion of CO ₂ to CO, %		61
2. Methanol Synthesis with Recycle		
catalyst, m ³	80	80
feed gas, m ³ /h	150 000	127 000
recycle gas, m ³ /h	620 000	506 000
produced methanol, tons/h	37.01	47.87
H ₂ O in raw methanol, wt %	38.2	15.3
specific methanol productivity, tons/m ³ _{cat} ·h	0.46	0.6
carbon conversion to methanol, %	69	89

Table 2. A Value in the Downstream Gas of the RWReactor Depending on the Reaction Conditions When CO₂ and H₂ Are Fed to the Reactor^a

temp	pressure	R					
		0	1	2	3	5	infinity
400	3	0.6369	0.5538	0.5056	0.4725	0.4293	0.2638
	30	0.6070	0.5057	0.4432	0.3986	0.3371	0.0350
500	3	0.5296	0.4350	0.3824	0.3477	0.3038	0.1541
	30	0.5030	0.3858	0.3206	0.2773	0.2215	0.0161
600	3	0.4148	0.3179	0.2676	0.2360	0.1970	0.0842
	30	0.3922	0.2858	0.2319	0.1931	0.1485	0.0090
700	3	0.3413	0.2489	0.2035	0.1758	0.1435	0.0558
	30	0.3217	0.2225	0.1730	0.1425	0.1054	0.0058
800	3	0.2682	0.1863	0.1482	0.1258	0.1004	0.0357
	30	0.2518	0.1646	0.1239	0.0999	0.0727	0.0036

^a Temperature, °C; pressure, atm; H₂/CO₂ = 3; R, recycled gas/produced gas (mole ratio).

of the possible sources for CO₂ separation might be the integrated gasification and combined cycle (IGCC). To mitigate CO₂ emission, hydrogen should not be produced from fossil fuels. Because of the second thermodynamic law, hydrogen productions from fossil fuels will lead to the net increase of CO₂ emission. Therefore, the possible energy source for H₂ production might be hydraulic, solar, or nuclear energy. The hydrogen production cost would be a crucial factor for the CAMERE process to be a practical one to produce methanol from CO₂. The

Table 3. Material Balance (mol/h) of the CAMERE Process

	CO	CO ₂	H ₂	H ₂ O	CH ₃ OH
Feed	0.000 00	2.300 50	7.762 65	0.000 00	0.000 00
Mix1-out	0.933 86	2.892 42	11.996 15	0.005 30	0.000 00
Reactor1-out	2.340 36	1.488 23	10.578 90	1.412 47	0.000 00
Water-out	0.000 00	0.001 16	0.000 00	1.399 23	0.000 00
Separator-out	2.340 36	1.487 07	10.578 90	0.013 24	0.000 00
Syn-gas	1.404 22	0.892 24	6.347 35	0.007 94	0.000 00
Split-stream	0.936 14	0.594 83	4.231 56	0.005 30	0.000 00
Mix-out	2.829 28	1.635 72	13.576 70	0.008 80	0.000 00
Reactor-out	2.045 15	1.094 56	10.360 40	0.550 29	1.342 43
Sep2-liq	0.000 57	0.025 54	0.001 68	0.549 07	1.319 80
Sep2-vap	2.044 58	1.069 02	10.358 70	0.001 23	0.022 63
Tee-out2	0.613 37	0.320 71	3.107 62	0.000 37	0.006 79
Tee-out1	1.431 20	0.748 31	6.251 10	0.000 85	0.015 84
H ₂ O	0.000 00	0.000 00	0.000 00	0.549 05	0.095 99
Methanol	0.000 57	0.025 54	0.001 68	0.000 01	1.223 81

Table 4. Battery Limit of the Equipment and Operation Costs for the Methanol Production Processes^a

process	equipment cost	operation cost (unit: \$1,000.00)
methanol from natural gas	26 269	2805
methanol from coal	30 130	53568
methanol by a high-pressure process	21 588	9607
CAMERE process	16 100	4134

^a Material costs are not involved in this estimation.

economical viability of the CAMERE process could be a milestone for the reduction of greenhouse effects.

4. Conclusions

It has been troublesome to develop an efficient and economical process to give methanol from CO₂ hydrogenation. The catalysts developed so far were not satisfactory in terms of activities and stabilities. We reported the CAMERE process for the reduction of CO₂. In the CAMERE process, the low productivity of the catalyst was overcome by the combination of the RWReactor with the methanol synthesis reactor. The alignment of two reactors reduced the recycle gas volume, minimizing the purge gas volume. The yield of methanol increased 29% as the purge gas volume decreased in the CAMERE process. An active and stable catalyst for the CAMERE process was developed. It had the composition

of Cu/ZnO/ZrO₂/Ga₂O₃ (5:3:1:1) and showed high activity and stability with the feed of high CO₂ concentration. The system was optimized, and the battery limits for the investment and operation cost were evaluated. The CAMERE process was compared with the commercial methanol synthesis processes from natural gas and coal. The overall aspects of the CAMERE process showed a strong possibility to produce methanol from the CO₂ hydrogenation economically.

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